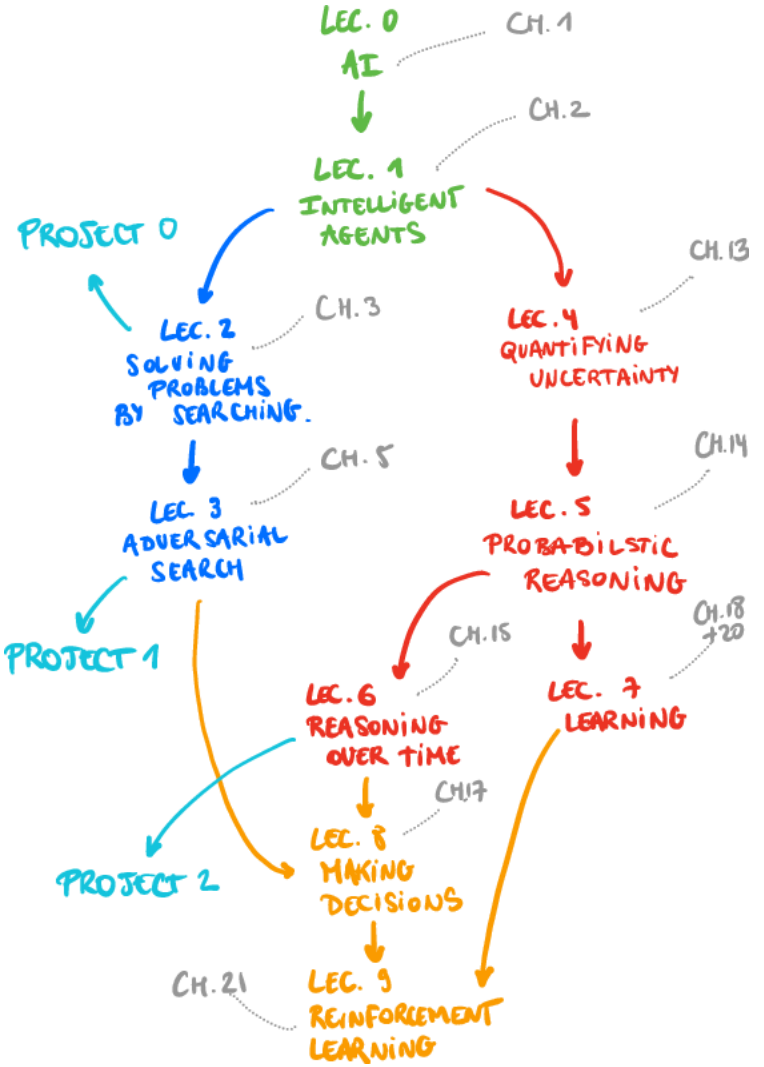
Introduction to Artificial Intelligence

Lecture 4: Quantifying uncertainty

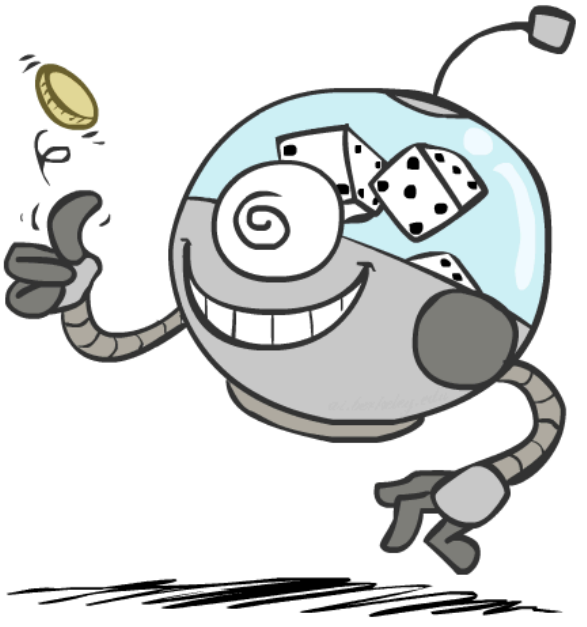
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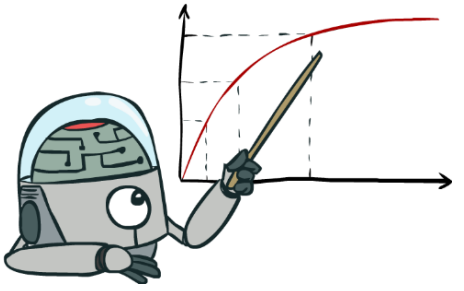

LEC. 0
AI
CH. 1
CH. 2
LEC. 1
INTELLIGENT AGENTS
PROJECT 0
CH. 3
LEC. 2
SOLVING PROBLEMS BY SEARCHING.
CH. 13
LEC. 4
QUANTIFYING UNCERTAINTY
CH. 5
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PROBABILSTIC REASONING
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LEC. 6
REASONING OVER TIME
CH. 18 +20
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LEARNING
CH. 17
LEC. 8
MAKING DECISIONS
PROJECT 2
CH. 21
LEC. 9
REINFORCEMENT LEARNING

Today

- Random variables
- Probability distributions
- Independence
- The Bayes' rule



> Do not overlook this lecture!



Principle of maximum expected utility

From Lecture 1, a rational agent chooses the action of **highest expected utility**, given what it has perceived,

$$a_t = \arg\max_{a \in \mathcal{A}} \mathbb{E}\left[V(s_{1:T}) \mid e_{1:t}, a_t = a\right],$$

where $V : \mathcal{S}^* \to \mathbb{R}$ scores the states $s_{1:T}$, and the average is over the states the agent may reach, following $P(s_{t+1} \mid s_t, a_t)$ and $P(e_t \mid s_t)$.

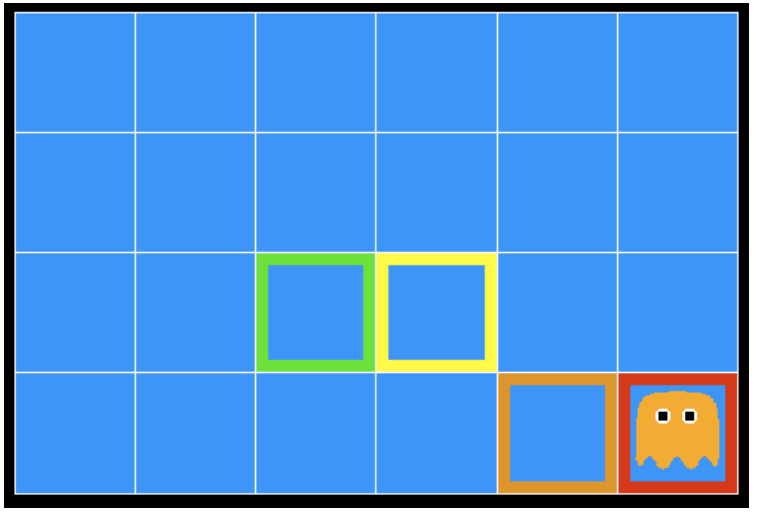
> What are these probabilities, and how do we reason with them?

Quantifying uncertainty

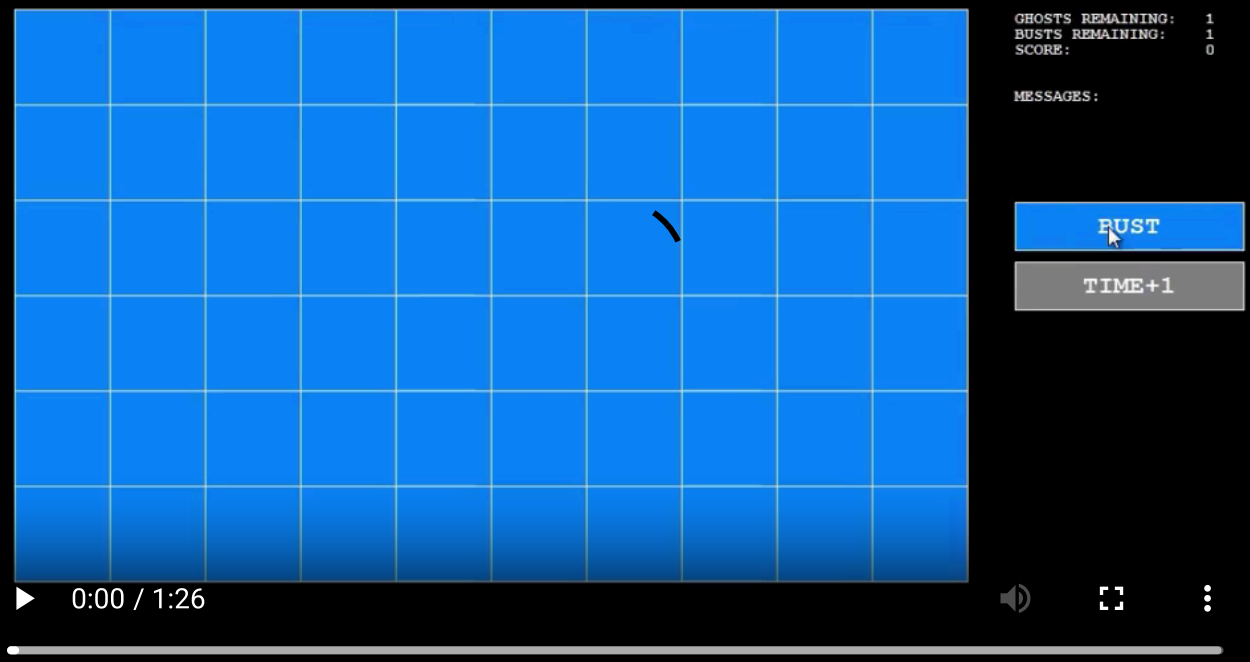
A ghost is **hidden** in the grid somewhere.

Sensor readings tell how close a square is to the ghost:

- On the ghost: red
- 1 or 2 away: orange
- 3 away: yellow
- 4+ away green



Sensors are **noisy**, but we know the probability values $P(\text{color} \mid \text{distance})$, for all colors and all distances.



Credits: CS188, UC Berkeley.

Uncertainty

General setup:

- **Observed** variables or evidence: agent knows certain things about the state of the world (e.g., sensor readings).
- **Unobserved** variables: agent needs to reason about other aspects that are uncertain (e.g., where the ghost is).
- (Probabilistic) **model**: agent knows or believes something about how the observed variables relate to the unobserved variables.

> **Probabilistic reasoning** provides a framework for managing our knowledge and beliefs.

Probabilistic assertions

A **probabilistic assertion** attaches a number to a proposition,

$$P(\text{ghost in cell } [3, 2]) = 0.02,$$

as a degree of belief, not as a claim about the truth of the proposition.

The agent holds beliefs rather than certainties for two reasons:

- **ignorance**: it does not observe the state, and its models of the world are approximate;
- **laziness**: listing every case and every exception would cost more than it is worth.

Frequentism vs. Bayesianism

What does a probability value represent? Take $P(\text{ghost in cell } [3, 2]) = 0.02$.

- The objectivist, **frequentist** view: a property of the world, the fraction of games in which the ghost sits in that cell.
- The subjectivist, **Bayesian** view: the degree of belief of the agent that the ghost is in that cell, given the readings it has taken.

The agent plays a single game, with no long run to count, so we read probabilities as beliefs in this course: a new reading changes the number, not the ghost.

Kolmogorov's axioms

Begin with a set Ω, the **sample space**. Its elements $\omega \in \Omega$ are the **sample points**, or possible worlds.

A **probability space** is a sample space equipped with a probability function P that assigns a number $P(\omega)$ to each sample point, such that

- $0 \leq P(\omega) \leq 1$ for all $\omega \in \Omega$;
- $\sum_{\omega \in \Omega} P(\omega) = 1$.

Events

An **event** is a set of sample points $A \subseteq \Omega$, and its probability is the mass it collects,

$$P(A) = \sum_{\omega \in A} P(\omega).$$

Equivalently, in Kolmogorov's terms,

- $P(A) \geq 0$ for all events;
- $P(\Omega) = 1$;
- $P(A \cup B) = P(A) + P(B)$ for disjoint events.

It follows that

$$P(\neg a) = 1 - P(a), \qquad P(a \vee b) = P(a) + P(b) - P(a \wedge b).$$

Example

- Ω = the 6 possible rolls of a die.
- ω_i (for $i = 1, ..., 6$) are the sample points, each corresponding to an outcome of the die.
- Assignment P for a fair die:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

- "3 or more" is the event $A = \{3, 4, 5, 6\}$, of probability

$$P(A) = P(3) + P(4) + P(5) + P(6) = \frac{2}{3},$$

hence $P(\neg A) = 1 - P(A) = \frac{1}{3}$ for "2 or less".

Random variables

- A **random variable** is a function $X : \Omega \to D_X$ from the sample space to a domain D_X of outcomes.
- "$X = x$" is the event $\{\omega \in \Omega : X(\omega) = x\}$, so P induces a **probability distribution** over X,

$$P(X = x) = \sum_{\{\omega : X(\omega) = x\}} P(\omega).$$

- The events we care about are the **assignments** of random variables, such as $X_1 = x_1, \ldots, X_n = x_n$.

Example:

On the die, "odd" is the random variable

$$\mathrm{Odd} : \Omega \to \{\mathrm{true}, \mathrm{false}\}, \qquad \mathrm{Odd}(\omega) = (\omega \bmod 2 = 1),$$

whose distribution is

$$P(\mathrm{Odd} = \mathrm{true}) = P(1) + P(3) + P(5) = \frac{1}{2}, \qquad P(\mathrm{Odd} = \mathrm{false}) = \frac{1}{2}.$$

Notations

- Random variables are named in upper case, X, Y, $\mathrm{Weather}$; their values in lower case, x, y, sun.
- D_X is the domain of X, so that $x \in D_X$ is one of its values.
- $P(X = x)$ is a **number**, abbreviated $P(x)$ when clear from context.
- $\mathbf{P}(X)$ is the **distribution** of X, i.e. the table of the numbers $P(X = x)$ for all $x \in D_X$, and not a single value.

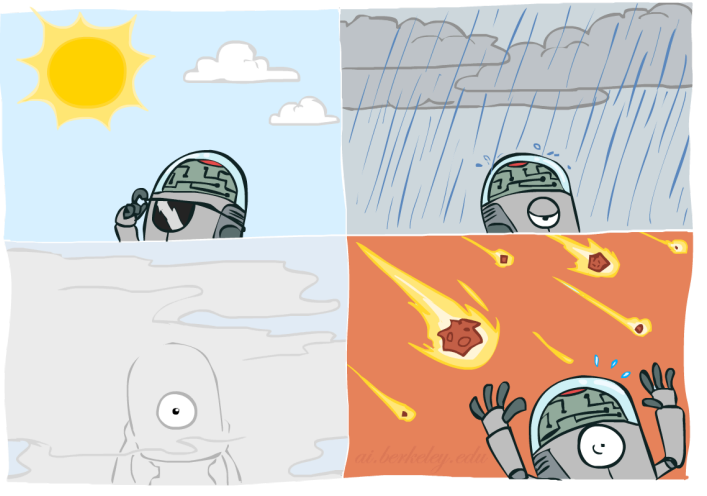
Probability distributions

Probability mass function

For a discrete variable X, the probability distribution $\mathbf{P}(X)$ is the list of the values $P(X = x)$ for all $x \in D_X$, known as the **probability mass function**.

$\mathbf{P}(W)$

W	$P(W)$
sun	0.6
rain	0.1
fog	0.3
meteor	0.0



Joint distributions

A **joint** probability distribution over a set of random variables $X_1, ..., X_n$ specifies the probability of each (combined) outcome:

$$P(X_1 = x_1, ..., X_n = x_n) = \sum_{\{\omega : X_1(\omega)=x_1, ..., X_n(\omega)=x_n\}} P(\omega)$$

$$\mathbf{P}(T, W)$$

T	W	$P(T, W)$
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Over n boolean variables, the table has 2^n entries, hence $2^n - 1$ free numbers.

Marginal distributions

The **marginal distribution** of a subset of random variables is obtained by summing the joint distribution over the values of the other variables (the **sum rule**),

$$P(x) = \sum_y P(x, y), \qquad \mathbf{P}(X) = \sum_y \mathbf{P}(X, Y = y).$$

$\mathbf{P}(T, W)$

T	W	$P(T, W)$
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$\mathbf{P}(T)$

T	$P(T)$
hot	0.5
cold	0.5

$P(t) = \sum_w P(t, w)$

$\mathbf{P}(W)$

W	$P(W)$
sun	0.6
rain	0.4

$P(w) = \sum_t P(t, w)$

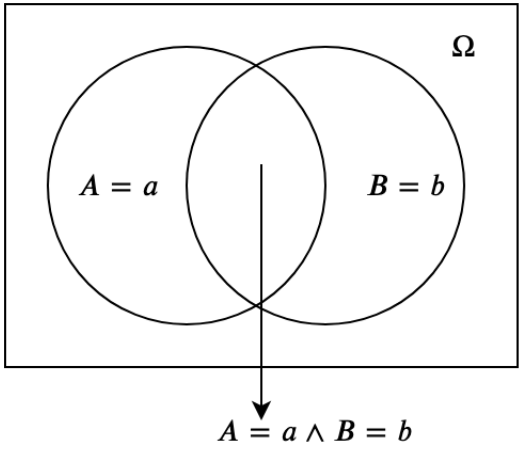
Intuitively, marginal distributions are sub-tables which eliminate variables.

Conditional distributions

The **conditional probability** of a given b, for $P(b) > 0$, is the ratio of the probability of a and b to the probability of b,

$$P(a \mid b) = \frac{P(a, b)}{P(b)}.$$

Indeed, observing $B = b$ rules out all those possible worlds where $B \neq b$, leaving a set whose total probability is just $P(b)$. Within that set, the worlds for which $A = a$ satisfy $A = a \wedge B = b$ and constitute a fraction $P(a, b)/P(b)$.



Conditional distributions are probability distributions over some variables, given **fixed** values for others. Writing $\mathbf{P}(X \mid Y)$ denotes them all at once: one distribution $\mathbf{P}(X \mid Y = y)$ for each value y of Y.

$\mathbf{P}(T, W)$

T	W	$P(T, W)$
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$\mathbf{P}(W \mid T = \text{hot})$

W	$P(W \mid \text{hot})$
sun	0.8
rain	0.2

$\mathbf{P}(W \mid T = \text{cold})$

W	$P(W \mid \text{cold})$
sun	0.4
rain	0.6

Product rule

Rearranging the definition of the conditional probability,

$$P(a, b) = P(a \mid b)P(b), \qquad \mathbf{P}(A, B) = \mathbf{P}(A \mid B)\mathbf{P}(B),$$

the second form standing for one equation per value of A and B.

Summing it over the values of B gives the **law of total probability**,

$$P(a) = \sum_b P(a \mid b)P(b).$$

$\mathbf{P}(W)$

W	$P(W)$
sun	0.8
rain	0.2

$\mathbf{P}(D \mid W)$

D	W	$P(D \mid W)$
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

$\mathbf{P}(D, W)$

D	W	$P(D, W)$
wet	sun	?
dry	sun	?
wet	rain	?
dry	rain	?

Chain rule

More generally, any joint distribution can always be written as an incremental product of conditional distributions:

$$P(x_1, x_2, x_3) = P(x_1)P(x_2 \mid x_1)P(x_3 \mid x_1, x_2)$$

$$P(x_1, ..., x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, ..., x_{i-1})$$

In the same way, $\mathbf{P}(X_1, ..., X_n) = \prod_{i=1}^{n} \mathbf{P}(X_i \mid X_1, ..., X_{i-1})$.

Independence

The joint distribution has d^n entries. The chain rule writes it as a product of conditional distributions,

$$P(x_1, ..., x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, ..., x_{i-1}),$$

but this alone saves nothing: the last factor, $P(x_n \mid x_1, ..., x_{n-1})$, still has d^n entries.

Independence and **conditional independence** state that some of these factors do not depend on all their conditions. Each dropped condition divides the size of a factor by d, which is what makes probabilistic models **scale up**.

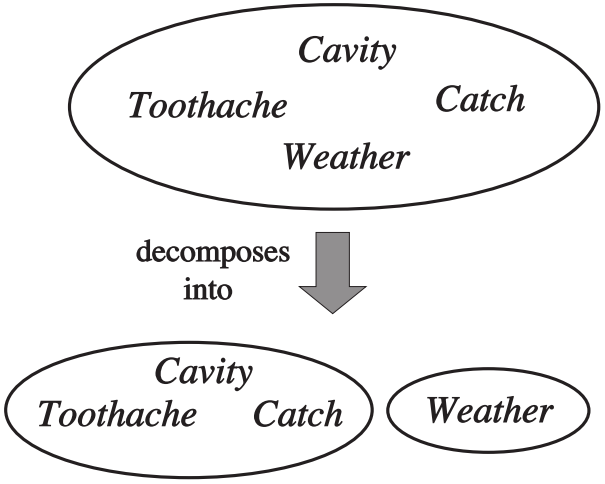
Independence

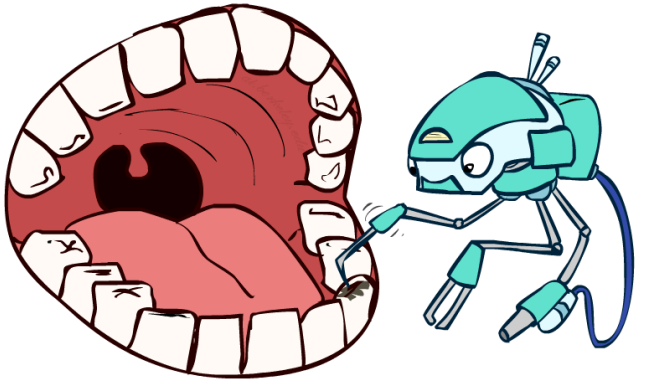
A and B are **independent**, denoted $A \perp B$, iff

$$P(a, b) = P(a)P(b) \quad \text{for all } a \in D_A, b \in D_B.$$

Equivalently, knowing one says nothing about the other: $P(a \mid b) = P(a)$ whenever $P(b) > 0$, and $P(b \mid a) = P(b)$ whenever $P(a) > 0$.

Example 1:





$$P(\text{toothache}, \text{catch}, \text{cavity}, \text{weather})$$
$$= P(\text{toothache}, \text{catch}, \text{cavity})P(\text{weather})$$

The original 32-entry table reduces to one 8-entry and one 4-entry table (assuming 4 values for Weather and boolean values otherwise).

Example 2:

For n independent coin flips, the joint distribution can be fully **factored** and represented as the product of n 1-entry tables.

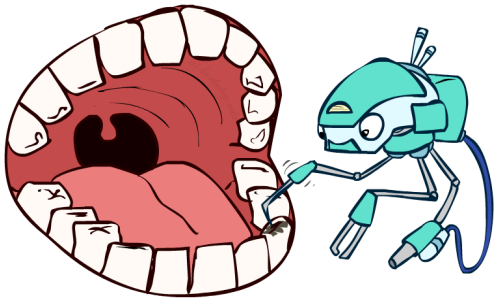
- $2^n \rightarrow n$

Conditional independence

A and B are **conditionally independent** given C, denoted $A \perp B \mid C$, iff

$$P(a, b \mid c) = P(a \mid c)P(b \mid c) \quad \text{for all } a \in D_A, b \in D_B, c \in D_C \text{ with } P(c) > 0.$$

Equivalently, once c is known, one says nothing more about the other:
$P(a \mid b, c) = P(a \mid c)$ and $P(b \mid a, c) = P(b \mid c)$, whenever these are defined.



Example 1:

Assume three random variables $\mathbf{Toothache}$, $\mathbf{Catch}$ and $\mathbf{Cavity}$.

$\mathbf{Catch}$ is conditionally independent of $\mathbf{Toothache}$, given $\mathbf{Cavity}$. Therefore, we can write:

$$\begin{aligned}
&P(\mathbf{toothache}, \mathbf{catch}, \mathbf{cavity}) \\
&= P(\mathbf{toothache} \mid \mathbf{catch}, \mathbf{cavity}) P(\mathbf{catch} \mid \mathbf{cavity}) P(\mathbf{cavity}) \\
&= P(\mathbf{toothache} \mid \mathbf{cavity}) P(\mathbf{catch} \mid \mathbf{cavity}) P(\mathbf{cavity})
\end{aligned}$$

In this case, the representation of the joint distribution reduces to $2 + 2 + 1$ independent numbers (instead of $2^n - 1$).

Example 2 (Naive Bayes):

More generally, from the product rule, we have

$$P(\text{cause}, \text{effect}_1, ..., \text{effect}_n) = P(\text{effect}_1, ..., \text{effect}_n \mid \text{cause})P(\text{cause})$$

Assuming the effects are **conditionally independent given the cause**, it comes:

$$P(\text{cause}, \text{effect}_1, ..., \text{effect}_n) = P(\text{cause}) \prod_i P(\text{effect}_i \mid \text{cause})$$

This probabilistic model is called a **naive Bayes** model.

- The complexity of this model is $O(n)$ instead of $O(2^n)$ without the conditional independence assumptions.
- Naive Bayes can work surprisingly well in practice, even when the assumptions are wrong.

The Bayes' rule

Causal and diagnostic directions

A model is usually given in the **causal** direction: a prior on the cause, and the distribution of its effects given the cause, as in the naive Bayes model. For the ghost and its sensors, $\mathbf{P}(G)$ and $\mathbf{P}(R_{i,j} \mid G)$.

Inference asks the **diagnostic** question, $\mathbf{P}(G \mid r_{i,j})$: the cause, given the effects that were observed.

Bayes' rule turns one into the other, and follows from the two factorizations of the product rule.

Study the next slide. **Twice**.

The Bayes' rule

The product rule defines two ways to factor the joint distribution of two random variables.

$$P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$$

Therefore,

$$P(a \mid b) = \frac{P(b \mid a)P(a)}{P(b)}.$$



- $P(a)$ is the prior belief on a.
- $P(b)$ is the probability of the evidence b.
- $P(a \mid b)$ is the posterior belief on a, given the evidence b.
- $P(b \mid a)$ is the conditional probability of b given a. Depending on the context, this term is called the likelihood.

$$P(a \mid b) = \frac{P(b \mid a)P(a)}{P(b)}$$



The Bayes' rule is the **foundation** of many AI systems.

Example 1: diagnostic probability from causal probability.

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause})P(\text{cause})}{P(\text{effect})}$$

where

- $P(\text{effect} \mid \text{cause})$ quantifies the relationship in the **causal** direction.
- $P(\text{cause} \mid \text{effect})$ describes the **diagnostic** direction.

Let S=stiff neck and M=meningitis. Given $P(s \mid m) = 0.7$, $P(m) = 1/50000$, $P(s) = 0.01$, it comes

$$P(m \mid s) = \frac{P(s \mid m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

Example 2: Ghostbusters, revisited

- A random variable G gives the ghost location, and a random variable $R_{i,j}$ the color read at the cell $[i, j]$.
- We start from a uniform **prior** $\mathbf{P}(G)$ over the locations.
- We know the sensors, i.e. the **sensor model** $\mathbf{P}(R_{i,j} \mid G)$.
  - e.g., $P(R_{1,1} = \textbf{yellow} \mid G = [1, 1]) = 0.1$.
- Readings are conditionally independent given the ghost location, $R_{i,j} \perp R_{i',j'} \mid G$.

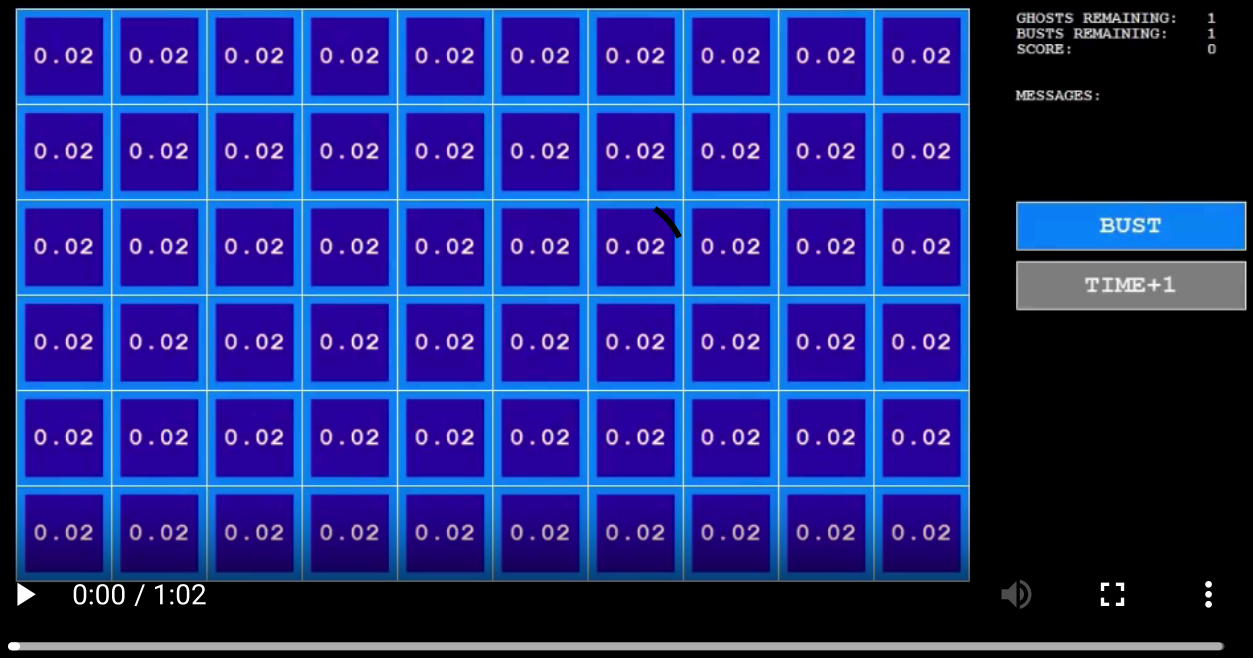
Once the reading r_1 is observed, Bayes' rule gives the **posterior** over the locations,

$$\mathbf{P}(G \mid r_1) = \frac{\mathbf{P}(r_1 \mid G)\mathbf{P}(G)}{P(r_1)} = \frac{1}{Z}\mathbf{P}(r_1 \mid G)\mathbf{P}(G).$$

Each new reading updates the posterior of the previous ones, which acts as its prior,

$$\mathbf{P}(G \mid r_{1:k+1}) = \frac{1}{Z}\mathbf{P}(r_{k+1} \mid G)\mathbf{P}(G \mid r_{1:k}),$$

since the readings are conditionally independent given G.


0.02
GHOSTS REMAINING: 1
BUSTS REMAINING: 1
SCORE: 0
MESSAGES:
BUST
TIME+1
0:00 / 1:02

Summary

- Uncertainty is **inescapable**, and **probability theory** is the framework we use to represent it: numbers attached to propositions, measuring the agent's beliefs rather than the world.
- The joint distribution answers every query, by selecting, marginalizing and normalizing, but it holds d^n entries.
- **Independence** and **conditional independence** factor it into smaller tables.
- **Bayes' rule** turns a causal model into a diagnostic belief, and each posterior is the prior of the next observation.

Recap

Sample space	Ω, of possible worlds ω
Axioms	$0 \leq P(\omega) \leq 1, \quad \sum_{\omega \in \Omega} P(\omega) = 1$
Event	$A \subseteq \Omega, \quad P(A) = \sum_{\omega \in A} P(\omega)$
Random variable	$X : \Omega \to D_X$, with $X = x$ the event $\{\omega : X(\omega) = x\}$
Distribution	$\mathbf{P}(X)$, the numbers $P(x)$ for $x \in D_X$
Joint distribution	$\mathbf{P}(X_1, \ldots, X_n)$, the numbers $P(x_1, \ldots, x_n)$
Conditional probability	$P(a \mid b) = P(a, b)/P(b)$, for $P(b) > 0$
Conditional distribution	$\mathbf{P}(X \mid Y)$, one $\mathbf{P}(X \mid y)$ for each $y \in D_Y$

Complement	$P(\neg a) = 1 - P(a)$
Union	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Product rule	$P(a, b) = P(a \mid b)P(b) = P(b \mid a)P(a)$
Sum rule, total probability	$P(x) = \sum_y P(x, y) = \sum_y P(x \mid y)P(y)$
Chain rule	$P(x_{1:n}) = \prod_i P(x_i \mid x_{1:i-1})$
Bayes' rule	$P(a \mid b) = P(b \mid a)P(a)/P(b)$
Independence, $X \perp Y$	$P(x, y) = P(x)P(y)$
Conditional independence, $X \perp Y \mid Z$	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$

The end.